

Omais Shafiq

Senior Structural & Design Engineer | Structural, Transportation & Computational Engineering

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Professional Summary

Senior Structural and Design Engineer with **6+ years of industry experience** spanning structural design, transportation infrastructure, and construction supervision across Pakistan and Malaysia. Proven track record managing and delivering multi-million-dollar projects for national highway authorities, urban mobility programmes, and government development initiatives. Dual expertise in **structural design** (RC frames, foundations, steel) and **road asset management** (RAMS, HDM-4, safety audits, FWD), complemented by advanced computational skills in ETABS, SAP2000, C#, and MATLAB. PhD candidate in structural engineering with 10 peer-reviewed publications — bringing research-grade rigour directly to engineering practice.

Education

- 2024–2027 **Doctor of Philosophy — Structural Engineering**, *Universiti Malaysia Pahang Al-Sultan Abdullah (UMPSA)*, Malaysia
 - Thesis: AI-integrated seismic fragility assessment and performance-based structural design.
 - Supervisor: Dr. Saffuan Bin Wan Ahmad. Research outputs: 5 journal papers submitted/under review (4 in Q1 venues).
- 2020–2024 **Master of Science — Structural Engineering**, *University of Engineering and Technology (UET)*, Peshawar, Pakistan
 - Thesis: Development of Fragility Curves for Single-Span Small Bridges in Peshawar (Conference Paper Published).
- 2015–2019 **Bachelor of Science — Civil Engineering**, *University of Engineering and Technology (UET)*, Peshawar, Pakistan
 - Thesis: Achieving High-Strength Concrete in a Challenging Environment (Published — GRIN Verlag, 2019).

Core Technical Competencies

Structural Design & Analysis

- Structural design of RC frames, foundations, slabs, beams, columns, and culverts to international codes (ACI, BS, AASHTO); experienced with ETABS and SAP2000 for building and bridge analysis.
- Developed *Conc* — a production desktop tool for RC beam/slab capacity checks and column interaction diagrams; validated steel frame designs for Australian truck loading.
- Seismic assessment and performance-based design; advanced knowledge of nonlinear analysis, fragility evaluation, and limit-state engineering.

Transportation & Road Asset Management

- National highway RAMS reporting (HDM-4, NHA network); road safety audit to international standards on major corridors including N-55 Indus Highway.
- Hands-on operation of Profilometer and Falling Weight Deflectometer (FWD) for pavement condition assessment; feasibility studies and technical sanctions for urban mobility schemes.

- Developed *Flexpave* — flexible pavement design software to AASHTO 2007 (C#, WPF, .NET).

Computational Engineering & Software Development

- Proficient in C#, MATLAB, .NET, and C++; built multiple production-grade desktop engineering applications deployed to practising engineers.
- ETABS API automation for parametric structural modelling and nonlinear analysis at scale; AutoCAD and Civil3D for design drafting and infrastructure planning.

Professional Experience

- 2022–2026 **Design Engineer**, *CEC Minconsult*, Pakistan, 4 years
- **Led delivery of transportation projects individually valued at over \$1 million USD**, reporting directly to the Principal Transportation Engineer — overseeing scope, compliance, and technical outputs end-to-end.
 - Compiled the **RAMS 2023 national reporting dataset** for the NHA road network under the CEC/PID partnership, informing asset maintenance policy across a major highway portfolio.
 - Prepared **Technical Sanctions for four Urban Mobility Projects** in Sahiwal and Sialkot; contributed to feasibility studies for RRDP and KPRAP.
 - Executed Road Safety Audit on **N-55 Indus Highway** (Shikarpur–D.I. Khan corridor) under KPRAP; conducted Social Impact Assessment surveys across Khyber Pakhtunkhwa.
- 2022–2023 **Structural Engineer**, *CEC Minconsult*, Pakistan, 8 months
- Reviewed and **fully redesigned the Women’s Business Development Centre, Kohat** — structural review, updated drawings, and compliance with KP Cities Improvement Project (KPCIP) standards.
 - Delivered **foundation designs for multiple KPCIP buildings** and culvert designs for RRDP and KPRAP road schemes.
- 2021–2022 **Site Engineer**, *Pakhtunkhwa Highway Authority*, Pakistan, 1 year
- Managed on-site execution of highway works — coordinated site staff, set and tracked daily production targets, and served as the direct interface between the Resident Engineer and contractor.
 - Identified and reported construction discrepancies, maintaining quality compliance on active highway corridors.
- 2019–2020 **Trainee Civil Engineer**, *Pakhtunkhwa Highway Authority*, Pakistan, 6 months
- Operated **Profilometer and Falling Weight Deflectometer (FWD)** for network-level pavement condition surveys; processed data through **HDM-4** to support road maintenance policy decisions.
- 2020–Present **Freelance Structural Engineer & Software Developer**, *Multiple Clients*, 4+ years
- Developed **Conc** — a production-grade desktop application for RC beam/slab capacity checks and column interaction diagrams (C#, WPF, .NET); deployed to practising engineers.
 - Developed **Flexpave** — flexible pavement design software to AASHTO 2007 (C#, WPF, .NET).
 - Built FEM-based analysis tools for masonry structures (MATLAB); delivered the EIDA seismic assessment pipeline for research clients.
- 2024–2025 **Visiting Lecturer**, *University of Engineering and Technology (UET)*, Peshawar, 1 year
- Delivered undergraduate courses in Structural Analysis, Dynamics, and Transportation Engineering; designed coursework and supervised laboratory experiments.

Publications

- [1] Shafiq, O., Wan Ahmad, S., Ali, S. M., Waqas, M., Khan, G., & Kamarudin, M. A. A. (2025). Fragility Curves Development for Single Span Slab Bridges: A Case Study for Peshawar, Pakistan. *Journal of Advanced Research in Applied Mechanics*, 133(1), 177–190. doi:10.37934/aram.133.1.177190 [Scopus]
- [2] Syed Mustorpha, S. I., Wan Ahmad, S., Amzar Kamarudin, M. A., Alsakkaf, H. A., & Shafiq, O. (2025). Advancements in Earthquake Research and Seismic Risk Analysis: An In-Depth Review Centered on Malaysia. *Journal of Advanced Research in Applied Sciences and Engineering Technology*, 63(2), 196–206. doi:10.37934/araset.63.2.196206 [Q2]
- [3] Kamarudin, M. A. A., Wan Ahmad, S., Alsakkaf, H. A., Syed Mustorpha, S. I., & Shafiq, O. (2025). The Assessment of Tall Building Structure Performance Due to Seismic Effect in Malaysia. *Journal of Advanced Research Design*, 129(1), 112–129. doi:10.37934/ard.129.1.112129 [Q4]

- [4] Waqas, M., Wan Ahmad, S., Shafiq, O., Ishaq, F., Laissy, M. Y., Adnan, A., & Al Sadoon, Z. A. (2025). Seismic Fragility Characterization of Grid-Like Frame Structures. *Construction*, 5(2). doi:10.15282/construction.v5i2.13231 [Q3]
- [5] Omair Shafiq, Abdul Jabbar, Ghufraan Ullah. *Achieving High Strength Concrete in Challenging Areas* (2019). Munich: GRIN Verlag. ISBN (eBook): 9783346037336 [Book]
- [6] Waqas, M., Wan Ahmad, S., Shafiq, O., & Talha, M. (2025). Advances in Earthquake Mitigation in Malaysia: A Review of Global Context and Local Progress. *Jurnal Teknologi*. (Accepted / In Press) [Q3]
- [7] Shafiq, O., Wan Ahmad, S., Arif, A. I. B. M., Ishaq, F., Khattak, M. M. H., Khan, G., Waqas, M., & Kamarudin, M. A. A. (2025). Factors Affecting Seismic Fragility of RC Grid-Like Frame Structures: A Parametric Meta-Regression Approach. *Structures*. (Revision Submitted — Round 1; MS No. STRUCTURES-D-25-09760) [Q1]
- [8] Shafiq, O., Wan Ahmad, S., Arif, A. I. B. M., Khan, G., Laissy, M. Y., Faizan, M., & Adnan, A. B. (2026). Probabilistic Seismic Demand Modeling of RC Grid-Like Frames: Assessing Response Sensitivity Across Parametric Structural Configurations. *Construction*. (Under Review) [Q3]
- [9] Shafiq, O., Wan Ahmad, S., & Iqbal, M. (2025). Parametric Sensitivity of the Seismic Demand Scaling Exponent to Design Variability and Soil–Structure Interaction in Structurally Similar RC Bare Frame Systems. *Structures*. (With Editor) [Q1]
- [10] Shafiq, O., Wan Ahmad, S., Khan, G., Khattak, M. M. H., & Ishaq, F. (2025). Transparent ML Framework for Seismic Fragility Derivation: A Closed-Form Dimensionless Approach. *Structures*. (Submitted) [Q1]
- [11] Shafiq, O., Wan Ahmad, S., Jintao, H., & Talha, M. (2025). An Automated Extended Incremental Dynamic Analysis Pipeline for High-Throughput Fragility Data Generation of RC Moment-Frame Buildings with Soil–Structure Interaction in ETABS. *MethodsX*. (Submitted) [Q1]

Skills

Programming	C#	.NET	C++	MATLAB	Visual Studio
Engineering Software	ETABS	SAP2000	AutoCAD	Civil3D	HDM-4